

CHAPTER – 1

ACCIDENTS IN INDIA

Introduction

This is the seventh edition of the report after the comprehensive revision of the proformae for data collection of 'Accidental Deaths & Suicides in India' was done in 2014 in consultation with States/UTs Police and others stakeholders.

An accident is an unplanned event that results in injury, deaths, damage to property or some other losses. For statistical purpose, Accidental Deaths have been classified as deaths that could be due to forces/factors of the nature which have been termed as

'Accidental Deaths due to Forces of Nature' or could be due to deliberate or negligent conduct of human beings, which is termed as 'Other causes of Accidental Deaths'.

The rate of accidental deaths (per lakh of population) has decreased in 2020 as compared to 2019. The total number of accidental deaths in 2020 is 3,74,397. [LIST-1.1]. The incidence and rate of accidental deaths during 2010 – 2020 is presented in Table-1.1.

LIST-1.1
Percentage Change in Population, Number and Rate of Accidental Deaths over the Corresponding Previous Year during 2016 to 2020

Year	Mid-Year Projected Population* (In Lakh ⁺)	Percentage Change in Population over the Previous Year	Total Number of Accidental Deaths	Percentage Change in Accidental Deaths over the Previous Year	Rate of Accidental Deaths	Percentage Change in Rate of Accidental Deaths over the Previous Year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2016	12739.9	1.2	418221	1.2	32.8	0.0
2017	13091.6 [#]	2.8	396584	-5.2	30.3	-7.6
2018	13233.8 [#]	1.1	411824	3.8	31.1	2.6
2019	13376.1 [#]	1.1	421104	2.3	31.5	1.3
2020	13533.9 ^{\$}	1.2	374397	-11.1	27.7	-12.1

* Source: The Registrar General of India.

Source: Report of the Technical Group on Population Projections (November, 2019), National Commission on Population, Ministry of Health & Family Welfare.

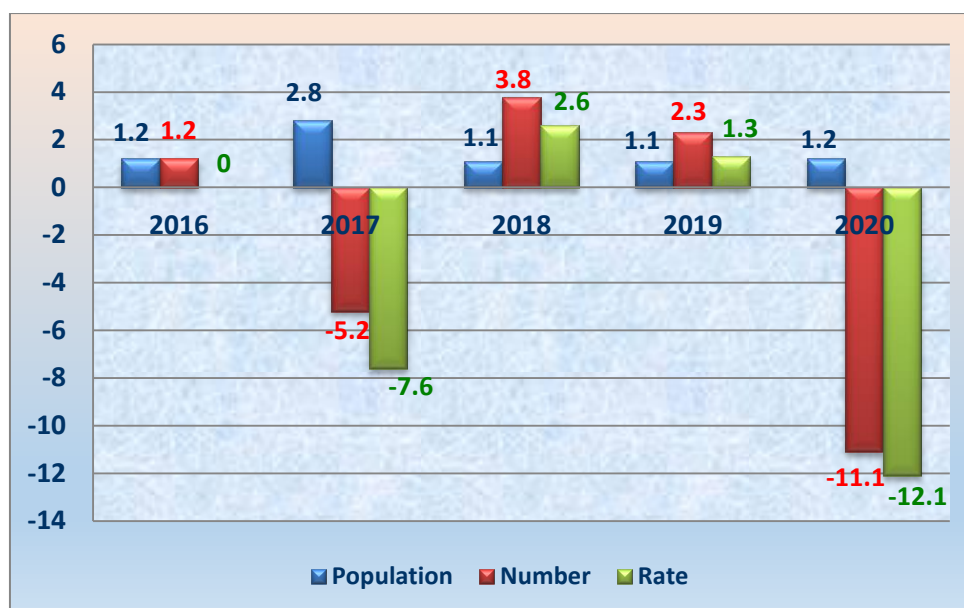
\$ Source: Report of the Technical Group on Population Projections (July, 2020), National Commission on Population, Ministry of Health & Family Welfare

+ One Lakh = 0.1 Million = 100,000

As per data provided by States/UTs

FIGURE-1.1

Percentage Change in Population, Number and Rate of Accidental Deaths during 2016 – 2020 (Over Corresponding Previous Year)



- As per data provided by States/UTs.

Out of 3,74,397 accidental deaths, 7,405 (2.0%) deaths were due to forces of nature and 3,66,992 (98.0%) deaths were due to other causes.

There was an decrease of 9.1% (from 8,145 deaths to 7,405 deaths) in deaths due to forces of nature and decrease of 11.1% (from 4,12,959 deaths to 3,66,992 deaths) in deaths due to other causes during 2020 as compared to 2019 **LIST-1.2 and Table 1.3.**

LIST-1.2

Number of Accidental Deaths due to Forces of Nature & Other Causes (2016 to 2020)

Year	Forces of Nature		Other Causes		Total Accidental Deaths	
	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2016	8,684	-17.4	4,09,537	1.6	4,18,221	1.2
2017	7,143	-17.7	3,89,441	-4.9	3,96,584	-5.2
2018	6,891	-3.5	4,04,933	4.0	4,11,824	3.8
2019	8,145	18.2	4,12,959	2.0	4,21,104	2.3
2020	7,405	-9.1	3,66,992	-11.1	3,74,397	-11.1

- As per data provided by States/UTs.

FIGURE-1.2
STATE/UT – WISE ACCIDENTAL DEATHS DURING 2020



• As per data provided by States/UTs.

Accidental Deaths

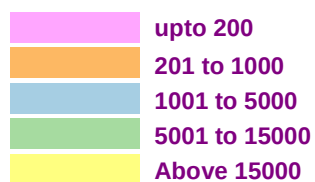
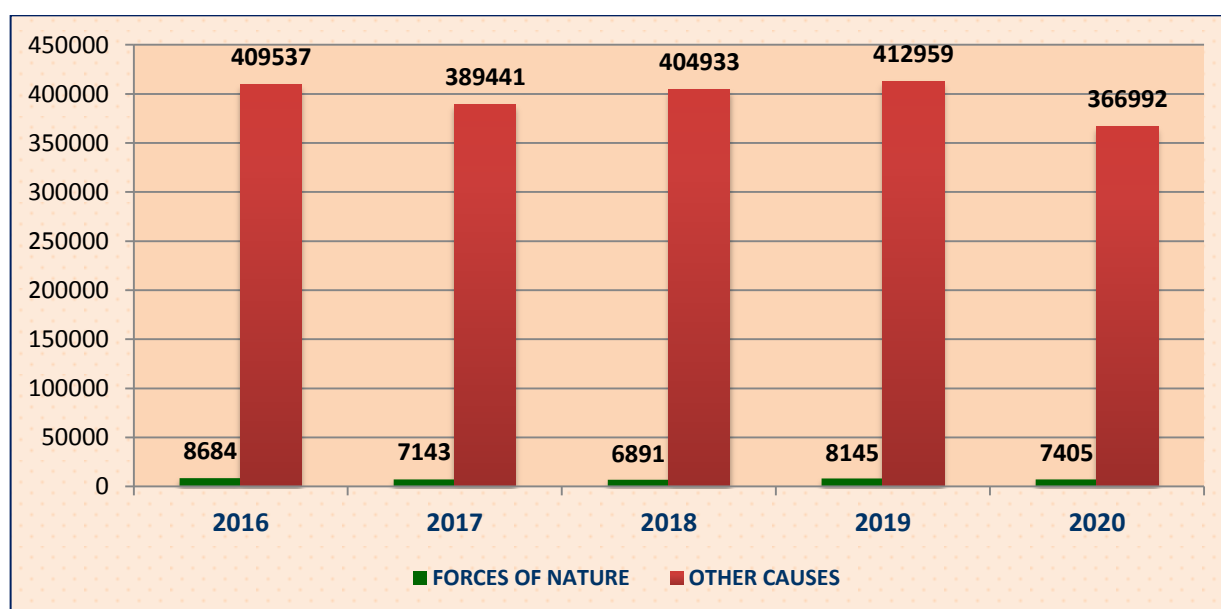


FIGURE-1.3

Total Number of Accidental Deaths due to Forces of Nature and Other Causes during 2016-2020



- As per data provided by States/UTs.

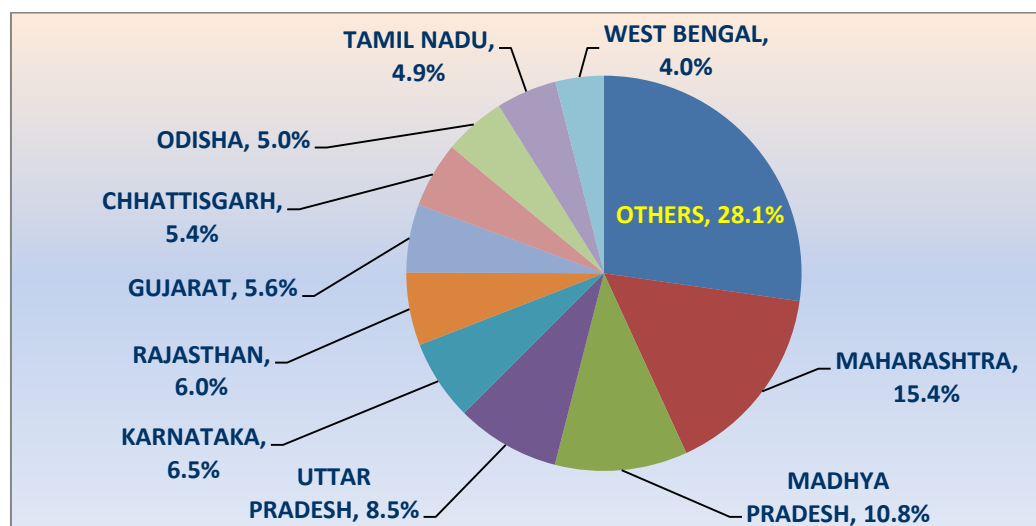
Number & Rate of Accidental Deaths in States, UTs and Cities

Maharashtra with a population share of 9.1% has reported highest number of accidental deaths (57,806), contributing nearly one-sixth (15.4%) of total accidental deaths reported in the country. Uttar Pradesh, the most

populous State in the country with population share of 16.9% has accounted for (31,691) 8.5% of the total accidental deaths in the country. The other States having higher percentage share in total accidental deaths were Madhya Pradesh (10.8%), Karnataka (6.5%), Rajasthan (6.0%), Gujarat (5.6%), Chhattisgarh (5.4%), Odisha (5.0%), Tamil Nadu (4.9%) and West Bengal (4.0%) [Table-1.2].

FIGURE-1.4

Percentage Share of Major States in Accidental Deaths during 2020



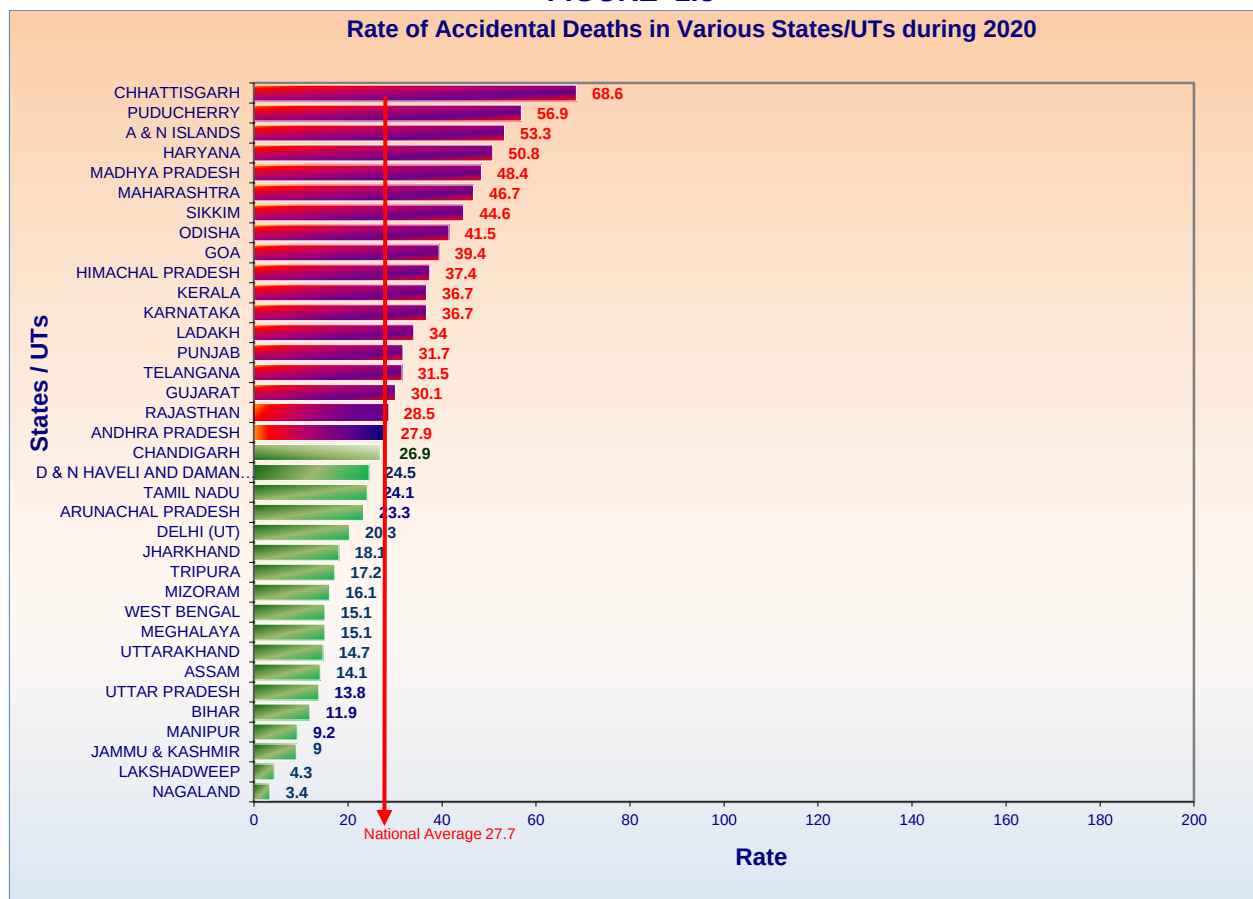
- As per data provided by States/UTs.

Other States/UTs include Andhra Pradesh, Arunachal Pradesh, Assam, Bihar, Goa, Haryana, Himachal Pradesh, Jharkhand, Kerala, Manipur, Mizoram, Meghalaya, Nagaland, Punjab, Sikkim, Telangana, Tripura, Uttarakhand, Chandigarh, A & N Island, D&N Haveli and Daman & Diu, Delhi, Jammu & Kashmir, Ladakh, Lakshadweep and Puducherry

The highest **rate of accidental deaths** was reported from Chhattisgarh (68.6) followed by Puducherry (56.9), A&N Islands (53.3), Haryana (50.8), Madhya Pradesh (48.4) and Maharashtra (46.7) against the national average of 27.7. Eighteen (18) out of 36

States/UTs have reported higher rates of 'Accidental Deaths' as compared to All India Average of 27.7 deaths per one lakh of population [**Figure – 1.5**].

FIGURE–1.5



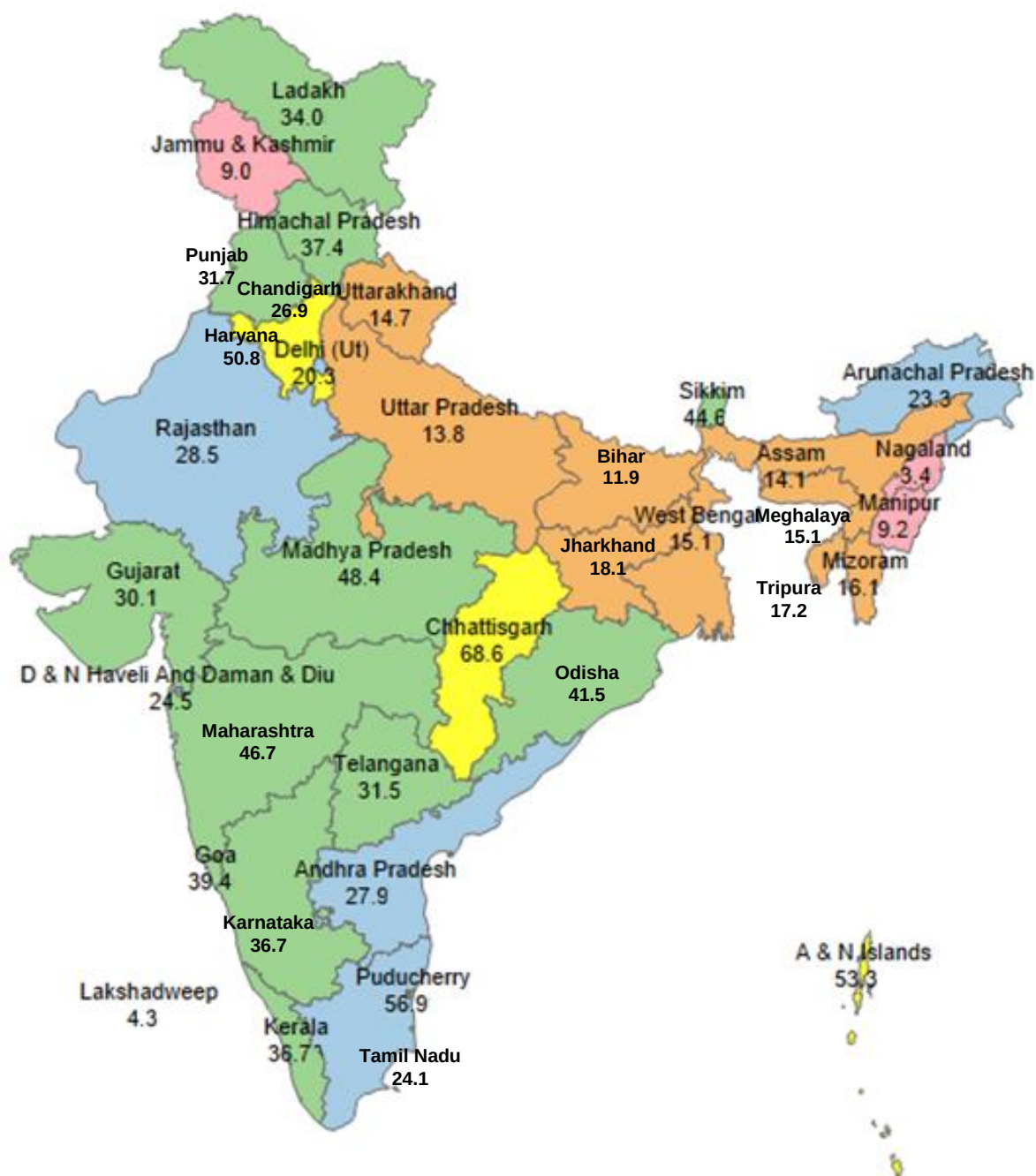
- As per data provided by States/UTs.

Arunachal Pradesh has reported the highest percentage increase (28.7%) in total accidental deaths in the year 2020 as compared to 2019 (from 275 in 2019 to 354 in 2020), followed by Nagaland (17.5%), Odisha (14.8%), Mizoram (14.0%) and A&N Islands (7.6%).

Lakshadweep has reported a decline of 40.0% in total accidental deaths in the year 2020 as compared to 2019 followed by Goa (23.1%), Chandigarh & Rajasthan (22.0% each) and Uttar Pradesh (21.9%). [**Table-1.3**].

Gender-wise analysis reveals that male and female constitute 80.5% and 19.5% of total victims respectively. Age group of majority of the victims was between 30 to 45 years. This group of people has accounted for 30.1% of all persons killed in accidents in the country during the year 2020. A total of 41,015 senior citizens (60 years & above of age) have died in various accidents during 2020 comprising 11.0%. [**Table-1.4**].

FIGURE-1.6
STATE/UT – WISE RATE OF ACCIDENTAL DEATHS DURING 2020



Note:

Rate of Accidental Deaths means Number of Accidental Deaths per 1,00,000 population.

- As per data provided by States/UTs.

Accidental Death Rate
All India Average 27.7

	upto 10
	10.1 to 20
	20.1 to 30
	30.1 to 50
	Above 50

Causes of Accidental Deaths

Accidental deaths have been grouped into two broad categories i.e. 'Deaths due to Forces of Nature' and 'Deaths due to Other Causes'. Classification of types and causes of accidental deaths has been done for statistical purpose only.

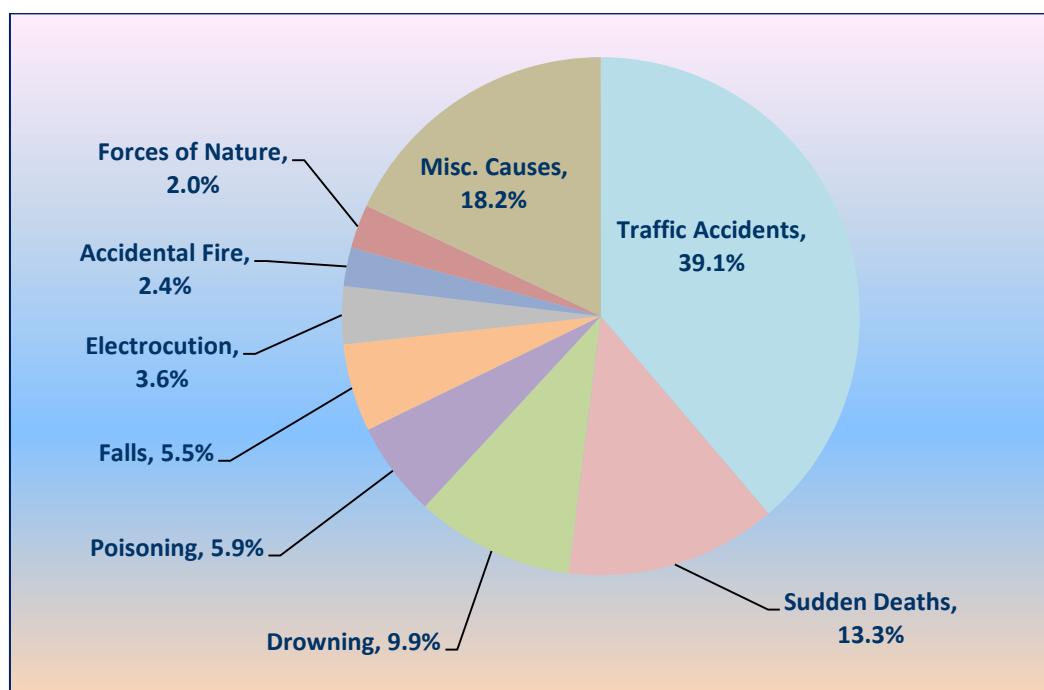
Under causes attributable to 'Forces of Nature', the deaths due to 'Forest Fire', 'Cyclone', 'Landslide', 'Tornado' and 'Flood' have increased during the year 2020 over the previous year 2019 whereas the deaths due to *inter-alia* 'Avalanche', 'Exposure to Cold', 'Earthquake', 'Heat/Sun Stroke', 'Lightning' and 'Torrential Rain' have decreased in 2020 over 2019.

The deaths due to Other Causes such as 'Air Crash', 'Suffocation', 'Stampede', 'Drowning', 'Sudden Deaths', 'Poisoning' and 'Electrocution' have increased whereas the deaths due to 'Traffic Accidents', 'Factory/Machine Accidents', 'Drug overdose', 'Collapse of Structure', 'Accidental Explosion', 'Accidental Fire', 'Firearms', 'Mines or Quarry Disaster', 'Deaths due to Consumption of Illicit/Poisonous Liquor', 'Falls', 'Deaths of Women during Pregnancy' and 'Killed by Animals' and have decreased in 2020 as compared to the previous year 2019.

There has been a decrease in number of traffic accidents deaths during 2020 (1,46,354 deaths) over during 2019 (1,81,113 deaths).

Cause-wise details of accidental deaths, its share and percentage variation during 2020 over 2019 are presented in **Table-1.0**.

FIGURE-1.7
Percentage Share of Various Major Causes of Accidental Deaths during 2020
(Forces of Nature and Other Causes)



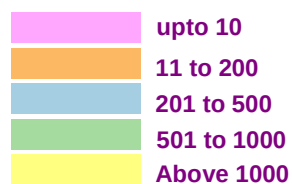
- As per data provided by States/UTs.

Misc. Causes include Collapse of Structure, Suffocation, Deaths of Women during Pregnancy, Killed by Animals, Deaths due to Consumption of Illicit/Poisonous Liquor, Accidental Explosion, Drug Overdose, Factory/Machine Accidents, Firearm, Mines or Quarry Disaster, Stampede, Air-Crash, Ship Accidents, Other than above & Unknown Causes.

FIGURE-1.8
STATE/UT – WISE DEATHS DUE TO FORCES OF NATURE DURING 2020



Accidental Deaths



- As per data provided by States/UTs.

Accidental Deaths due to Forces of Nature

Cause - wise accidental deaths due to forces of nature are presented in **Table-1.0**. Out of 7,405 accidental deaths due to forces of nature, 38.6% deaths due to 'Lightning' (2,862 deaths), 13.0% deaths due to 'Flood' (959 deaths), 10.5% deaths due to 'Exposure to Cold' (776 deaths) and 7.2% deaths due to 'Heat/Sun Stroke' (530 deaths) during the year 2020. 'Landslide' and 'Cyclone' caused 4.0% (295 deaths) and 0.5% (37 deaths) of total deaths due to forces of nature respectively.

The age group-wise distribution of accidental deaths due to forces of nature is given in **Table-1.5**. As per age-wise classification of deaths due to forces of nature, 6.7% of such victims were below 14 years of age (493 out of 7,405 deaths), 7.3% in the age-group of 14 years to below 18 years (540 deaths), 18.9% in the age-group of 18 years to below 30 years (1,398 deaths), 24.5% in the age-group of 30 years to below 45 years (1,816 deaths) and 21.1% victims were in age group 45 years to below 60 years (1,560 deaths) during 2020. Senior citizens (60 yrs. and above) constituted 21.6% (1,598 deaths) of the total such victims.

State/UT - wise and gender - wise distribution of accidental deaths in the country by forces of nature along with their share in total deaths is presented **Table-1.9**. 'Lightning' was the dominant factor and caused 91.1%, 90.1, 76.5% and 70.7%, of total such deaths in Chhattisgarh (246 out of 270 death), Tamil Nadu (64 out of 71), Madhya Pradesh (429 out of 561 deaths) and Jharkhand (336 out of 475 deaths) respectively. 54.7% (786 out of 1437 deaths) of deaths in Bihar was due to 'Flood'. 71.6% (53 out of 74 deaths) of death in Haryana were due to 'Exposure to Cold'. 'Heat/Sun Stroke' was the dominant factor and caused 57.6% (98 out of 170 deaths) of total accidental deaths due to forces of nature in Telangana. 'Landslide' was the dominant factor and caused 100.0%, 89.7%, 60.3% and 57.9% of total deaths in Mizoram (4 out of 4 deaths), Kerala (70 out of 78 deaths), Uttarakhand (44 out of 73 deaths) and Sikkim (44 out of 73 deaths) respectively during 2020.

Majority of accidental deaths due to 'Cyclone' were reported in West Bengal which accounted for 59.5% (22 out of 37 deaths) of total such deaths reported in the country during 2020. Majority of deaths due to 'Torrential Rain' were reported in Jharkhand which accounted for 23.3% (10 out of 43 deaths) of total such deaths in the country during 2020.

Accidental Deaths due to Other Causes

It is generally accepted that accidents due to 'Other Causes' are preventable and can be reduced by effective safety measures and safety consciousness.

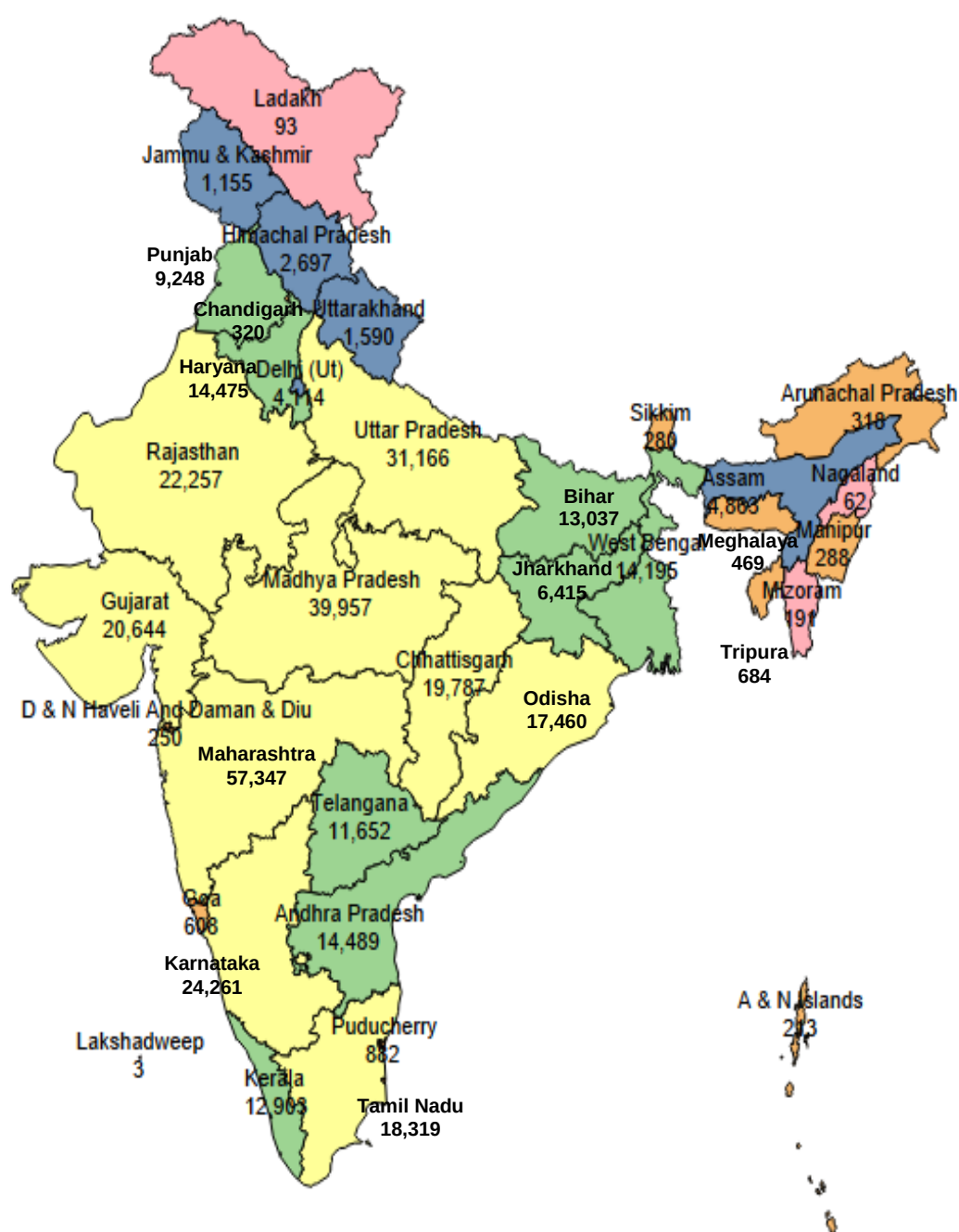
A total of 5,88,738 cases of Accidents due to 'Other Causes' were reported in the country which resulted in 3,66,992 deaths and 3,38,903 persons injured during 2020 [**Table-1.7**].

The cause-wise analysis of deaths revealed that majority of deaths were due to traffic accidents which accounted for 39.9% (1,46,354 out of 3,66,992 total deaths) of total deaths due to 'Other Causes' during the year 2020. Besides, 'Sudden Deaths', 'Drowning', 'Poisoning' and 'Falls' caused 13.6% (49,925 deaths), 10.1% (37,238 deaths), 6.1% (22,221 deaths) and 5.6% (20,579 deaths) of total deaths due to other causes respectively. [**Table-1.0**]

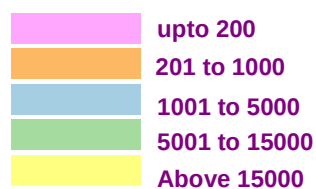
The age-group-wise distribution of accidental deaths in the country by various 'Other Causes' along with their share in total deaths is presented in **Table-1.8**. Majority of victims were in age groups 30 - below 45 years and 18 - below 30 years which accounted for 30.3% (1,11,058 deaths) and 26.0% (95,340 deaths) of total such deaths respectively.

State/UT-wise and gender-wise distribution of accidental deaths in the country by various other causes is presented in **Table-1.10**. Maharashtra with 9.2% of country's population has reported 15.6% (57,347 out of 3,66,992 deaths) of accidental deaths due to other causes. Madhya Pradesh, Uttar Pradesh, Karnataka, Rajasthan, Gujarat and Chhattisgarh accounted for 10.9%, 8.5%, 6.6%, 6.1%, 5.6% and 5.4% of total such deaths respectively during the year 2020.

FIGURE-1.9
STATE/UT – WISE ACCIDENTAL DEATHS DUE TO OTHER CAUSES
DURING 2020



Accidental Deaths



- As per data provided by States/UTs.

State/UT-wise and Cause-wise details on 'Accidents due to Other Cause' is presented in **Table-1.10**. State/UT-wise analysis reveals that Uttar Pradesh accounted for maximum deaths (21,156 out of 1,46,354) under traffic accidents. Maharashtra accounted for maximum deaths (15,245 out of 49,225) under 'Sudden Deaths'. Madhya Pradesh accounted for maximum deaths (5,779 out of 37,238) under 'Drowning'. Madhya Pradesh accounted for maximum deaths (4,611 out of 22,221) under 'Poisoning'.

A total of 931 incidents of consumption of illicit/spurious liquor caused 947 deaths in the country during 2020. States where various such deaths were reported include Madhya Pradesh (214) followed by Jharkhand (139 deaths) , Punjab (133 deaths), Karnataka (99 deaths) and Chhattisgarh (67 deaths).

Fire Accidents

Considering the number of fire accidents in the country and potential of the damage that can be caused by such accidents, the Bureau has also collected data on fire accidents according to places of occurrence for appropriate preventive strategies viz. 'Fire in School Buildings', 'Fire in Commercial Buildings', 'Fire in Residential / Dwelling Buildings', 'Fire in Government Buildings', 'Fire in Mines', 'Fire in Factories', 'Accidental Fire in

Trains' and 'Accidental Fire in Vehicles' [**Table-1.11**].

A total of 9,329 cases of fire accidents were reported in the country during 2020, showing a decrease of 15.5% as compared to 2019 (11,037 cases). 9,329 incidents of fire accidents caused injuries to 468 persons and 9,110 deaths during 2020. The cause-wise analysis of fire accidents revealed that 57.6% of total deaths (5,248 out of 9,110) were reported in residential/dwelling buildings during 2020 [**Table 1.12**].

Large number of cases of fire accidents were reported in Madhya Pradesh (1,430 out of 9,329), accounting for 15.3% of total such cases during 2020 [**Table-1.12**].

The State/UT-wise analysis of place of fire accidents revealed that 25 States & UTs have reported 50.0% or more deaths due to 'Fire in residential or dwelling building' during 2020. These States/UTs were Assam, Manipur, Mizoram, Sikkim, Tripura ,D&N Haveli and Daman Diu, Ladakh (100% each) and Uttarakhand (96.3%), Tamil Nadu (92.9%) .[**Table-1.12**].

Accidents in Cities

The term 'Mega City' refers to a city which has a population of 10 lakhs or more. As per the Population Census 2011, the status of mega city has been accorded to 53 cities.

A total of 50,512 accidental deaths were reported in 53 mega cities during 2020. Maximum numbers of accidental deaths (5,221 deaths) were reported in Mumbai accounting for 10.3% of total deaths, followed by Delhi City (3,994), Bengaluru (3,644), Pune (2,599), Nagpur (2,258) and Surat (2,119).

Rate of accidental deaths in these 53 mega cities was higher (31.4) than all India rate of accidental deaths (27.7). The rate of accidental deaths was highest in Raipur (92.1), followed by Rajkot (90.8), Nagpur (90.3), Faridabad (88.2), Aurangabad (81.8) and Vasai Virar (80.9) as compared to average rate for the cities (31.4) [Table-1.2].

Out of 50,512 accidental deaths in 53 mega cities; 311 and 50,201 deaths were due to 'Forces of Nature' and 'Other Causes' respectively during 2020. A decrease of 25.6% was observed in deaths due to forces of nature and an decrease of 17.7% was witnessed in deaths due to 'Other Causes' during 2020 over 2019 respectively. [Table-1.3]

Kolkata, the third largest city in terms of population (8.8% of the total population of all mega cities) among 53 cities, reported the low share of deaths in cities accounting for 0.9% of total accidental deaths in 53 mega cities during 2020. The comparative reporting of 'Accidental Deaths' during 2020 in the top four-metropolitan cities is presented in LIST-1.3.

Accidental Deaths due to Forces of Nature in Cities

A total of 311 deaths due to forces of nature were reported in these 53 mega cities during 2020. Among specified causes of accidents due to forces of nature, maximum deaths were reported due to 'Exposure to cold' followed by 'Heat/Sun stroke', accounting for 40.8% and 21.2% of total such accidental deaths. Due to 'Exposure to cold' maximum deaths were reported in Amritsar (47 deaths), Ludhiana (28 deaths) and Ranchi (9 deaths). Due to 'Heat/Sun stroke' deaths were reported in Ahmedabad (1 out of 2 death) and Hyderabad (1 out of 2 death) 50.0% (each), 42.9% deaths in Ludhiana (21 out of 49 death), 40.0% in Ghaziabad (2 out of 5 deaths) and 38.5% in Dhanbad (5 out of 13 deaths).

Accidental Deaths due to Other Causes in Cities

A total of 87,016 cases of 'Other' accidents were reported in 53 mega cities which caused injuries to 41,910 persons and 50,201 deaths during 2020. The cause-wise analysis of deaths due to other causes revealed that 26.7%, 24.8%, and 6.9% deaths in 53 cities were due to 'Sudden Deaths' (13,385), 'Traffic Accidents' (12,461), and 'Falls' (3,468) respectively.

Among 53 cities, Mumbai city (5,220) accounted for 10.4% of total deaths reported under 'Other' accidents followed by Delhi City (3,992) and Bengaluru (3,644) which accounted for 8.0% and 7.3 % respectively during 2020.

"Data on City-wise information is available on our website under 'Additional Tables for ADSI'"

LIST-1.3
Rate of Accidental Deaths in Metropolitan Cities during 2020

Sl. No.	City	Population (In Lakhs)	Share of Population In Cities (Percent)	Share of Deaths In Cities (Percent)	Rate (No. of Deaths Per One Lakh Population)
(1)	(2)	(3)	(4)	(5)	(6)
1.	Chennai	87.0	5.4	2.8	16.3
2.	Delhi	163.2	10.2	7.9	24.5
3.	Kolkata	141.1	8.8	0.9	3.4
4.	Mumbai	184.1	11.5	10.3	28.4

- As per data provided by States/UTs.
